

These practice MCQ (50) were written by SGU faculty and test some of the material taught in the following lectures:

Lipids and membrane transport (8), digestion and absorption (11), glycolysis, gluconeogenesis, regulation, blood glucose, fructose, galactose (10), PDH, TCA cycle, ETC, oxid. Phosph.. (7), glycogen metabolism and GSD (5), PPP and ROS (9)

Please take the exam in a row (50 MCQ in 75 min) and read afterwards the explanations.

Q# 1: You have isolated tissue that requires only pentose-phosphates but not NADPH for biochemical processes. The enzyme that would be active in the tissue is

- A. Transketolase
- B. Glucose 6-phosphate dehydrogenase
- C. 1,3 bisphosphoglycerate mutase
- D. Pyruvate dehydrogenase
- E. Pyruvate carboxylase

Q#2: Glucose 6-phosphate dehydrogenase variants are quite common in populations. Low activity of the enzyme glucose 6-phosphate dehydrogenase would result in most severe manifestations involving the

- A. Brain
- B. Muscle
- C. Liver
- D. Adipose tissue
- E. Erythrocyte

Q# 3: A 42 year-old man in prison is on a hunger strike for 5 days. Which of the following would be true regarding his liver metabolism?

- A. Glycogen phosphorylase is dephosphorylated
- B. Glycogen synthase is active
- C. Fructose 2,6 bisphosphate is absent
- D. Phosphofructokinase-1 is active
- E. Glucokinase is very active

Q# 4: A type I diabetic patient injects himself with 10 Units of insulin. The metabolic process observed after the injection of insulin is

- A. Activation of liver glycogen phosphorylase

- B. Phosphorylation of glycogen synthase
- C. Activation of adenylate cyclase
- D. Decreased activity of glucose 6-phosphatase
- E. Increased number of GLUT-4 on adipocytes

Q# 5: A 6 month-old girl has been brought by her parents to their family physician. Her blood glucose was 55 mg/dL (Normal: 60-90 mg/dL). Further investigation lead to the diagnosis of Hers disease. Which enzyme is deficient in this child?

- A. Muscle glycogen phosphorylase
- B. Glucose 6-phosphatase
- C. Branching enzyme in liver
- D. Debranching enzyme in liver
- E. Liver glycogen phosphorylase

Q#6: A patient is suspected to have a deficiency of the first enzyme of the pentose phosphate pathway beginning with glucose 6-phosphate. Which of the following would be affected in the liver of the individual?

- A. The conversion of glucose to galactose
- B. The conversion of glucose to pyruvate
- C. The conversion of oxidized glutathione to reduced glutathione
- D. The oxidation of NADH in the electron transport chain
- E. The conversion of fructose to glucose

Q#7: During newborn screening, a child was found to have elevated levels of galactose in blood and urine on the second day of birth. The process that is abnormal in this inborn disorder is

- A. Decreased conversion of glucose to galactose
- B. Decreased conversion of galactose to glucose
- C. Decreased conversion of fructose to glucose
- D. Increased accumulation of hydrogen peroxide
- E. Decreased conversion of lactose to galactose and glucose

Q# 8: A 50 year-old male has ingested 50 grams of glucose as part of the oral glucose tolerance test. 30 minutes after ingesting glucose, which of the following enzymes would be most active in the liver?

- A. Glycogen phosphorylase kinase
- B. Glucose 6-phosphatase
- C. Pyruvate carboxylase

- D. Fructose 1,6 bisphosphatase
- E. Phosphofructokinase-1

Q# 9: Oxidative phosphorylation is being studied in the laboratory. To an isolated mitochondrial preparation, 2,4 dinitrophenol is added. This leads in this experiment to

- A. Increased ATP formation from ADP
- B. Decreased translocation of ADP into the mitochondrial membrane
- C. Inhibition of electron transfer from complex I to complex III
- D. Transport of H^+ from the inter-membrane space to the mitochondrial matrix
- E. Inhibition of electron transport from complex IV to oxygen

Q# 10: Al Martini has hemolytic episodes on administration of sulfa drugs. Hemolysis in Al Martini is due to

- A. Decreased ATP generation by RBCs
- B. Insufficient reduced GSH levels
- C. Decrease in oxidized glutathione levels
- D. Insufficient oxygen carrying capacity of Hb
- E. Decreased levels of 2,3-bisphosphoglycerate

Q# 11: During a glucose tolerance test, 30 minutes after ingestion of 75 grams of glucose, which of the following metabolic changes are characteristic?

- A. Increased activity of glycogen phosphorylase
- B. Phosphorylation of glycogen synthase
- C. Increased levels of cAMP in hepatocyte
- D. Formation of fructose 2,6-bisphosphate
- E. Increased activity of phosphoenolpyruvate carboxykinase

Q#12: A 6 month-old child with a history of seizures is evaluated. The blood glucose level at the time of the seizure was 40 mg/dL. There is gross hepatomegaly. Injection of glucagon caused no elevation in blood glucose level. The biochemical defect in the child would most probably be a defect in

- A. Muscle glycogenolysis
- B. Liver glycogenolysis
- C. Conversion of galactose to glucose
- D. Conversion of fructose to glycolytic intermediates
- E. Muscle glycogenesis

Q# 13: A patient with a history of alcohol abuse is admitted to the hospital with symptoms of lethargy and an irregular heartbeat. Further tests find increased levels of lactic acid. What is the initial diagnosis? And what would be the initial treatment?

- A. Wernicke-Korsakoff syndrome / Oligomycin treatment
- B. Beriberi / Thiamine supplements
- C. Leigh syndrome / NSAIDs
- D. Ragged Red Fibers / amino acid supplements
- E. Early onset Alzheimer's disease / Biotin supplements

Q# 14: The electron transport chain can be inhibited at several stages, and the site of action of the inhibitor can be identified by the redox states of the electron carriers and acceptors. Which of the following is a complex III inhibitor?

- A. Cyanide
- B. Antimycin A
- C. Sodium azide
- D. Rotenone
- E. Amytal

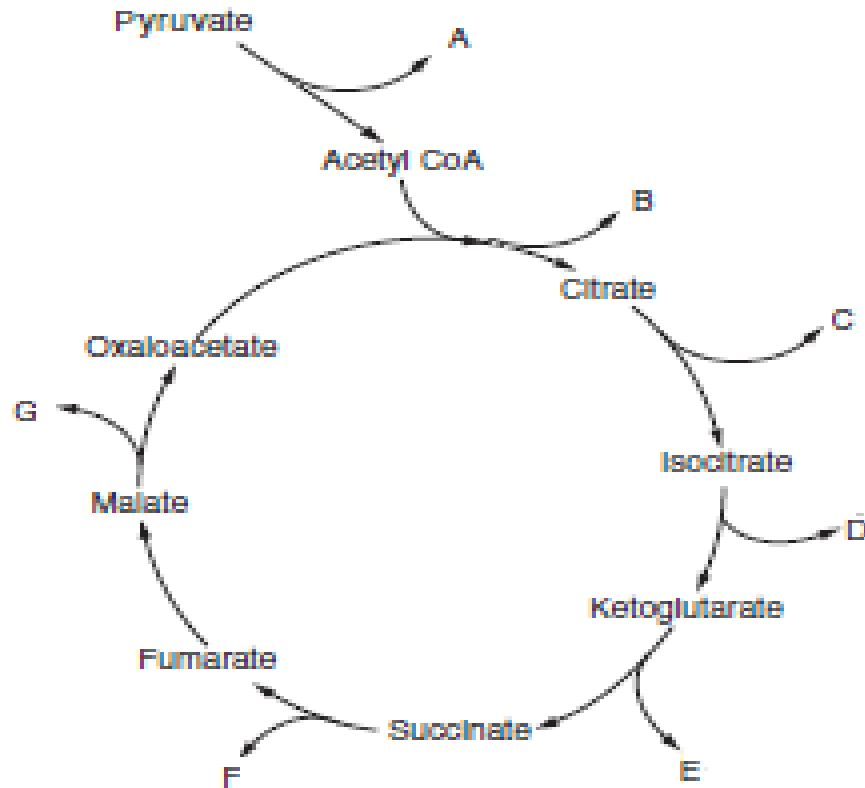
Q# 15: Hypoxia can result from anemia, pulmonary obstruction, high altitudes as well as binding of certain substances to hemoglobin. Which of the following substances can bind to the ferrous iron in the heme center of hemoglobin?

- A. 2,3-BPG
- B. carbon dioxide
- C. Cyanide
- D. Carbon monoxide
- E. Nitrogen gas

Q# 16: Many small molecules can disrupt ETC and oxidative phosphorylation. Which of the following would act by preventing the passage of electrons along the electron transport chain?

- A. 2,4 dinitrophenol
- B. Oligomycin
- C. Antimycin A
- D. Aspirin
- E. Atractyloside

Q# 17: A young girl comes to the medical center, and the consulting doctor suspects thiamine deficiency. Which steps of the pathways are likely to be affected?



- A. B and C
- B. A and G
- C. D and E
- D. A and E
- E. B and F

Q# 18: Gluconeogenesis and glycolysis share reversible steps but they contain also irreversible steps. Which of the following enzymes is used in gluconeogenesis but not in glycolysis?

- A. Pyruvate kinase
- B. Phosphofructokinase-1
- C. Fructose 1,6-bisphosphatase
- D. Aldolase B
- E. Glyceraldehyde 3-P dehydrogenase

Q# 19: The metabolite glucose 6-phosphate

- A. inhibits glycogen synthesis in the liver
- B. stimulates glycogen degradation in muscle
- C. is formed in muscle by glucokinase
- D. is cleaved in liver ER to glucose and Pi
- E. is changed to fructose 6-P during gluconeogenesis

Q# 20: Fructose 2,6-bisphosphate

- A. is formed in the committed step of glycolysis
- B. is cleaved to fructose 6-P during gluconeogenesis
- C. activates gluconeogenesis in kidney
- D. feed-forward activates pyruvate kinase
- E. is formed by the phosphorylated liver bifunctional enzyme

Q# 21: After high intake of ethanol, which of the following steps of gluconeogenesis in the liver is reduced due to high cytosolic NADH/NAD⁺ ratio?

- A. Formation of pyruvate from lactate
- B. Carboxylation of pyruvate
- C. Glucose 6-phosphatase
- D. Formation of malate from oxaloacetate in cytosol
- E. Transamination of alanine to pyruvate

Q# 22: Gluconeogenesis uses lactate as part of the Cori cycle. Which of the following enzymes is found in mitochondria and is needed for the eventual usage of this substrate?

- A. Glucose 6-phosphatase
- B. Pyruvate carboxylase
- C. PFK-2/Fruct 2,6BPase
- D. Pyruvate dehydrogenase
- E. Pyruvate kinase

Q# 23: Which enzyme generates glucose 1-P in the liver at insulin ruling?

- A. Glucokinase
- B. Glycogen phosphorylase
- C. Glycogen debranching enzyme
- D. Phosphoglucomutase
- E. Glycogen synthase

Q# 24: Which of the following enzymes contains calmodulin subunits?

- A. Glycogen debranching enzyme
- B. Glycogen phosphorylase kinase
- C. Glycogen phosphorylase
- D. Glycogen synthase
- E. Glycogen branching enzyme

Q# 25: In which glycogen storage disease do you find abnormal cytosolic glycogen structure with short branches (limit dextrin) and mild hypoglycemia in patients?

- A. Pompe disease
- B. Anderson disease
- C. McArdle syndrome
- D. Cori disease
- E. Hers disease

Q# 26: Reactive oxygen species (ROS) are a group that contains radicals and also non-radical compounds. Which of the following descriptions of superoxide is correct? Superoxide

- a. is a non-radical oxygen
- b. is formed by uptake of two electrons
- c. is formed in the ETC at CoQ level
- d. is degraded by NADPH oxidase
- e. is used as substrate for NO synthase

Q# 27: Hydrogen peroxide is continuously formed as by-products of aerobic metabolism and needs to be scavenged. Which of the following descriptions regarding hydrogen peroxide is correct? Hydrogen peroxide is

- a. scavenged by superoxide dismutase
- b. used to form hydroxyl radicals by catalase
- c. used as substrate for glutathione peroxidase
- d. is concentrated in the cell in mitochondria
- e. is a reactive oxygen species (ROS) that is a radical

Q#28: The healthy body has many ways to scavenge ROS with enzymes or with specific compounds. On the other hand, ROS can be formed on purpose for the defense against bacteria. Key enzymes in the oxygen-dependent system are NADPH oxidase and myeloperoxidase. Which of the following describes actions performed by myeloperoxidase?

- a. Generation of superoxide

- b. Use of hydrogen peroxide and Cl^- as substrates
- c. Formation of peroxynitrite from nitric oxide and superoxide
- d. Use of reduced glutathione
- e. Oxygen consumption leading to the respiratory burst

Q# 29: Hydrogen peroxide leads to the detrimental hydroxyl radical. Which of the following descriptions related to hydroxyl radicals is correct? Hydroxyl radical is

- a. formed from hydrogen peroxide and ferrous iron
- b. generated at the level of CoQ of the ETC
- c. scavenged by catalase
- d. formed by NADPH oxidase
- e. formed from water without radiation

Q# 30: The deficiency of which enzyme leads to recurrent infections with candida albicans?

- a. NADPH oxidase
- b. Superoxide dismutase
- c. Myeloperoxidase
- d. Catalase
- e. Inducible Nitric oxide synthase

Q# 31: Dependent on the polarity, we separate polar lipids and neutral lipids. Neutral lipids, like triacylglycerols (TAGs), are cleaved by lipases. One of the lipases that digest dietary TAGs is lingual lipase which

- a. needs colipase for its activity
- b. is acid-stable at the stomach proton concentration
- c. needs conjugated bile salts for its activity
- d. is deficient in patients with cystic fibrosis
- e. cleaves both, TAGs and phospholipids

Q# 32: Lipids are a major component of our normal diet and we have several digestive lipases. What is special of pancreatic lipase, often just named lipase? Pancreatic lipase

- a. cleaves dietary triacylglycerols to mainly free glycerol and fatty acids
- b. in serum can be an injury marker for pancreatitis
- c. cleaves mainly triacylglycerols with medium-chain fatty acids
- d. is proteolytically activated by trypsin
- e. is released as zymogen from the pancreas

Q# 33: During your clinical rotations, you come to learn about a family with a child who suffers from Cystic Fibrosis. Not only is the child treated with physical clearing of the lungs from the accumulating mucus, but also a special nutrition is of importance and part of the treatment. What is special about the digestion in cystic fibrosis patients? Patients with cystic fibrosis

- a. lack the glucose transporter SGLT-1
- b. are deficient in the digestion performed by pepsin
- c. should eat a diet low in lactose
- d. can normally digest TAGs with medium-chain fatty acids
- e. should eat a diet low in purine nucleotides

Q#34: Digestion of food starts in the mouth and after swallowing it continues in the stomach. The stomach performs the digestion of

- a. sucrose and lactose
- b. TAGs with long-chain fatty acids
- c. proteins to mainly peptides
- d. nucleic acids to bases and sugars
- e. phospholipids to fatty acids and lysophospholipids

Q#35: There are several hormones involved with the digestion of our normal food intake. Amongst them are gastrin, secretin and cholecystokinin (CCK). What is special about CCK? Cholecystokinin

- a. is a hormone formed in the pancreas
- b. activates trypsinogen to trypsin
- c. leads to release of mainly bicarbonate from the pancreas
- d. leads to release of protons into the stomach lumen
- e. leads to release of bile from the gallbladder

Q# 36: A 54-year-old woman complains to her physician of feeling night-blind while driving home late at night. It happened to her only recently, and she is worried about her health. Upon questioning, she tells her physician that she is since last month very strict with her diet, which is mostly fat-free. There is no change in her feces. What could most likely be the reason for her problems?

- a. Less uptake of the lipid-soluble vitamin A
- b. Lack of conjugated bile salts
- c. The deficiency of dietary cholesterol
- d. A diet that is too high in proteins
- e. A diet too high in purine nucleotides

Q# 37: Cholesterol is released by the liver into the bile. The solubility of cholesterol in bile is achieved by conjugated bile salts and phosphatidylcholine. Taurocholic acid is formed from cholic acid and taurine. This conjugated bile salt is

- a. synthesized in the gallbladder
- b. the main component of gallstones
- c. able to prevent gallstone formation
- d. grouped as secondary conjugated bile salt
- e. mainly released and excreted in feces

Q# 38: Digestive enzymes are very powerful enzymes that cannot only digest dietary proteins or phospholipids, but also cell components which are not meant to be digested. Proteolytic activation of digestive zymogens is often used. A key player in that respect is trypsin which proteolytically activates in the duodenum

- a. pepsinogen
- b. enteropeptidase
- c. chymotrypsinogen
- d. α -amylase
- e. pancreatic lipase

Q# 39: A 19-year-old Danish female developed severe diarrhea on a hiking trip. After her return, she consults a physician as she still suffers from continued abdominal pain with bloating and painful cramping. This takes place especially after eating sweets and drinking milk.

What could be most likely the reason for her problems?

- a. Deficiency of disaccharidases due to her severe diarrhea
- b. A strong genetic predisposition for primary lactose intolerance
- c. Congenital lactase deficiency
- d. Deficiency of secretin and cholecystokinin
- e. Inability to cleave lactose in the stomach

Q# 40: Most phospholipases are found inside of cells and are membrane-bound and tightly regulated. One of the exceptions is a soluble phospholipase A₂ that cleaves dietary phospholipids. This enzyme

- a. is the result of proteolytic activation by enteropeptidase
- b. has a pH optimum of 2
- c. is synthesized as prophospholipase in the stomach
- d. is eventually degraded by digestive proteases
- e. is released in its active form by the pancreas

Q# 41: The digestion of dietary carbohydrates is very much watched in programs that promote weight loss. One could target the enzymes that degrade complex sugars or one could target the uptake of glucose and galactose into intestinal mucosal cells. This uptake is performed by

- a. mainly facilitated transport
- b. GLUT-4
- c. GLUT-2
- d. Primary active transport
- e. Co-transport with sodium ions

Q# 42: Regulations allow changes in metabolism dependent on the changing needs of the cells or the total human body. Which of the following occurs in the liver when the blood glucose level is low and glucagon is ruling (low insulin/glucagon ratio)?

- a. Induction of key enzymes for gluconeogenesis
- b. Mobilization of GLUT-4

- c. Inhibition of adenylate cyclase
- d. Dephosphorylation of key liver enzymes of pathways
- e. Calmodulin activation by protein kinase A

Q#43: Membranes contain different transporters. Which of the following is an example of an ABC-transporter?

- a. SGLT
- b. GLUT-1
- c. CFTR
- d. GLUT-5
- e. Na^+/K^+ -ATPase

Q#44: Hepatocytes contain the specific glucose transporter GLUT-2. This transporter is

- a. Insulin-dependent
- b. A low-affinity transporter
- c. Only found in hepatocytes
- d. Mainly found in RBC
- e. Only used for uptake of glucose but not for its release from cells

Q#45: A 56-year-old man was seriously injured in an accident. He needs parenteral nutrition, which should include dietary essential fatty acids. Which of the following fatty acids is a dietary essential fatty acid of the ω -6 family?

- a. Palmitoleic acid
- b. Oleic acid
- c. Linoleic acid
- d. α -linolenic acid
- e. Arachidonic acid

Q#46: Digestive proteases have specific cleavage sites and work all together in the intestinal lumen in order to efficiently digest proteins to amino acids. Which amino acid side chain contains a positive charge and is (besides arginine residues) the cleavage site for trypsin?

- a. Asparagine
- b. Lysine
- c. Glutamine
- d. Tryptophan
- e. Histidine

Q#47: The plasma membrane separates the cell from its neighbor cells and the extracellular or interstitial fluid. The correct fluidity of the plasma membrane is regulated by

- a. Free cholesterol which buffers the fluidity changes of fatty acids during temperature changes
- b. The amount of glucose transporters and other proteins
- c. Arachidonic acid in phospholipids which will reduce the fluidity

- d. Palmitic acid which is more fluid than oleic acid
- e. Free cholesterol which increases the fluidity at the area of the polar head groups

Q#48: Phospholipids are components of polar lipids that often can be used to form biomembranes due to their amphipathic character. The group includes cardiolipin which contains totally four esterified fatty acids. What is special about cardiolipin? Cardiolipin is

- a. Exclusively found in the heart
- b. Found in high concentration in mitochondria
- c. A major component of all biomembranes
- d. Part of the glycocalyx
- e. A sphingophospholipid

Q#49: Premature babies are often treated with lung surfactant which

- a. Is a glycosphingolipid
- b. Is a plasmalogen
- c. Lines the alveoli of the lungs and prevents collapse
- d. Contains mainly protein and some phospholipids
- e. Contains two arachidonic acids in its structure

Q#50: The pyruvate dehydrogenase complex links aerobic glycolysis to the TCA cycle. Its regulation involves

- a. Activation by NADH
- b. Activation by acetyl CoA
- c. Inhibition by phosphorylation
- d. Inhibition by calcium ions
- e. Inhibition of PDH kinase by ATP